

大营销项目性能测试报告

01-测试目的

- 1. 优化代码，找bug；
- 2. 得出一组真实的TPS和RT数据

02-测试工具

jmeter+prometheus+grafana

03-测试环境

3.1 环境

购买4c8g三台服务器，带宽100M；

1. 一台： prometheus, grafana, jmeter
- ```
[root@iZ2zefck8g31o73sxh0mapZ ~]# docker ps
```

| CONTAINER ID | IMAGE                     | COMMAND                  | CREATED     | STATUS     | PORTS              |
|--------------|---------------------------|--------------------------|-------------|------------|--------------------|
| 5b4a2b4dba4e | grafana/grafana:10.2.0    | "/run.sh"                | 7 hours ago | Up 4 hours | 3000/tcp,          |
| na           |                           |                          |             |            |                    |
| ce066826aaea | bitnami/prometheus:2.47.2 | "/opt/bitnami/promet..." | 7 hours ago | Up 4 hours | 0.0.0.0:9090->9090 |
| theus        |                           |                          |             |            |                    |
| 9153436fd1ac | portainer/portainer       | "/portainer"             | 7 hours ago | Up 4 hours | 8000/tcp,          |
| iner         |                           |                          |             |            |                    |
2. 一台： mysql, phpmyadmin, redis, redis-admin, zookeeper, rabbitmq, canal-adapter, canal-server, elasticsearch, kibana, nacos

| <input type="checkbox"/> | Name          | State   | Icon | Query | Actions | Stack   | Image                            | Created             |
|--------------------------|---------------|---------|------|-------|---------|---------|----------------------------------|---------------------|
| <input type="checkbox"/> | canal-adapter | running |      |       |         | dev-ops | coopertools/canal-adapter:v1.1.6 | 2024-09-09 10:43:53 |
| <input type="checkbox"/> | canal-server  | running |      |       |         | dev-ops | canal/canal-server:v1.1.7        | 2024-09-09 10:43:53 |
| <input type="checkbox"/> | elasticsearch | running |      |       |         | dev-ops | elasticsearch:7.17.14            | 2024-09-09 10:43:48 |
| <input type="checkbox"/> | kibana        | running |      |       |         | dev-ops | kibana:7.17.14                   | 2024-09-09 10:43:53 |
| <input type="checkbox"/> | mysql         | healthy |      |       |         | dev-ops | mysql:8.0.32                     | 2024-09-09 10:43:48 |
| <input type="checkbox"/> | nacos         | running |      |       |         | dev-ops | nacos/nacos-server:v2.2.3-slim   | 2024-09-09 10:43:53 |
| <input type="checkbox"/> | phpmyadmin    | running |      |       |         | dev-ops | phpmyadmin:5.2.1                 | 2024-09-09 10:43:53 |
| <input type="checkbox"/> | portainer     | running |      |       |         | -       | portainer/portainer              | 2024-09-09 09:58:58 |
| <input type="checkbox"/> | rabbitmq      | running |      |       |         | dev-ops | rabbitmq:3.12.9                  | 2024-09-09 17:01:53 |
| <input type="checkbox"/> | redis         | healthy |      |       |         | dev-ops | redis:6.2                        | 2024-09-09 17:01:53 |
| <input type="checkbox"/> | redis-admin   | healthy |      |       |         | dev-ops | spryker/redis-commander:0.8.0    | 2024-09-09 10:43:53 |
| <input type="checkbox"/> | zookeeper     | running |      |       |         | dev-ops | zookeeper:3.9.0                  | 2024-09-09 10:43:48 |

3. 一台：nginx, big-market-app-01, big-market-app-02, big-market-front-app, xxl-job-admin

|                          |                      |         |  |  |  |         |                              |   |
|--------------------------|----------------------|---------|--|--|--|---------|------------------------------|---|
| <input type="checkbox"/> | big-market-app-01    | running |  |  |  | dev-ops | ben/big-market-app:4.0       | : |
| <input type="checkbox"/> | big-market-app-02    | running |  |  |  | dev-ops | ben/big-market-app:4.0       | : |
| <input type="checkbox"/> | big-market-front-app | healthy |  |  |  | dev-ops | ben/big-market-front-app:4.0 | : |
| <input type="checkbox"/> | nginx                | running |  |  |  | dev-ops | nginx:1.25.1                 | : |
| <input type="checkbox"/> | portainer            | running |  |  |  | -       | portainer/portainer          | : |
| <input type="checkbox"/> | xxl-job-admin        | running |  |  |  | dev-ops | xuxueli/xxl-job-admin:2.4.1  | : |

3.2 设置启动参数

```
server:
 port: 8091
 tomcat:
 threads:
 max: 800 # 设定处理客户请求的线程的最大数目，决定了服务器可以同时响应客
 # 户请求的数，默认200
```

```
min-spare: 100 # 初始化线程数,最小空闲线程数,默认是10
accept-count: 1000 # 等待队列长度
max-connections: 20000

thread:
 pool:
 executor:
 config:
 core-pool-size: 20
 max-pool-size: 50
 keep-alive-time: 5000
 block-queue-size: 5000
 policy: CallerRunsPolicy

数据库连接池
pool:
 pool-name: Retail_HikariCP
 minimum-idle: 15 #最小空闲连接数量
 idle-timeout: 180000 #空闲连接存活最大时间,默认600000 (10分钟)
 maximum-pool-size: 400 #连接池最大连接数,默认是10
 auto-commit: true #此属性控制从池返回的连接的默认自动提交行为,默认值: true
 max-lifetime: 1800000 #此属性控制池中连接的最长生命周期,值0表示无限生命周期,默认1800000即30分钟
 connection-timeout: 30000 #数据库连接超时时间,默认30秒,即30000
 connection-test-query: SELECT 1

容器中mysql的最大连接数修改成了2000
mysql -uroot -p

SHOW VARIABLES LIKE 'max_connections';

SET GLOBAL max_connections = 2000;
```

3.3 往raffle\_account中插入20w用户

|                                |    |            |        |      |      |      |      |      |      |   |
|--------------------------------|----|------------|--------|------|------|------|------|------|------|---|
| tables                         | 6  | liergou2   | 100301 | 100  | 86   | 100  | 86   | 100  | 86   | : |
| raffle_activity_account        | 7  | user003    | 100301 | 100  | 98   | 100  | 98   | 100  | 98   | : |
| raffle_activity_account_day    | 8  | xiaofuge26 | 100301 | 1    | 1    | 1    | 1    | 1    | 1    | : |
| raffle_activity_account_month  | 9  | ben1       | 100301 | 2000 | 2000 | 2000 | 2000 | 2000 | 2000 | : |
| raffle_activity_order_000      | 10 | ben2       | 100301 | 2000 | 2000 | 2000 | 2000 | 2000 | 2000 | : |
| raffle_activity_order_001      | 11 | ben7       | 100301 | 2000 | 2000 | 2000 | 2000 | 2000 | 2000 | : |
| raffle_activity_order_002      | 12 | ben8       | 100301 | 2000 | 2000 | 2000 | 2000 | 2000 | 2000 | : |
| raffle_activity_order_003      | 13 | ben9       | 100301 | 2000 | 2000 | 2000 | 2000 | 2000 | 2000 | : |
| task                           | 14 | ben12      | 100301 | 2000 | 2000 | 2000 | 2000 | 2000 | 2000 | : |
| user_award_record_000          | 15 | ben13      | 100301 | 2000 | 2000 | 2000 | 2000 | 2000 | 2000 | : |
| user_award_record_001          | 16 | ben14      | 100301 | 2000 | 2000 | 2000 | 2000 | 2000 | 2000 | : |
| user_award_record_002          | 17 | ben15      | 100301 | 2000 | 2000 | 2000 | 2000 | 2000 | 2000 | : |
| user_award_record_003          | 18 | ben23      | 100301 | 2000 | 2000 | 2000 | 2000 | 2000 | 2000 | : |
| user_behavior_rebate_order_000 | 19 | ben24      | 100301 | 2000 | 2000 | 2000 | 2000 | 2000 | 2000 | : |
| user_behavior_rebate_order_001 | 20 | ben25      | 100301 | 2000 | 2000 | 2000 | 2000 | 2000 | 2000 | : |
|                                | 21 | ben26      | 100301 | 2000 | 2000 | 2000 | 2000 | 2000 | 2000 | : |
|                                | 22 | ben34      | 100301 | 2000 | 2000 | 2000 | 2000 | 2000 | 2000 | : |

04-测试场景

模拟不同用户抽奖多次。接口响应时间是200ms，线程梯度：50，100，150，200，250，300，350，400个线程，200次；

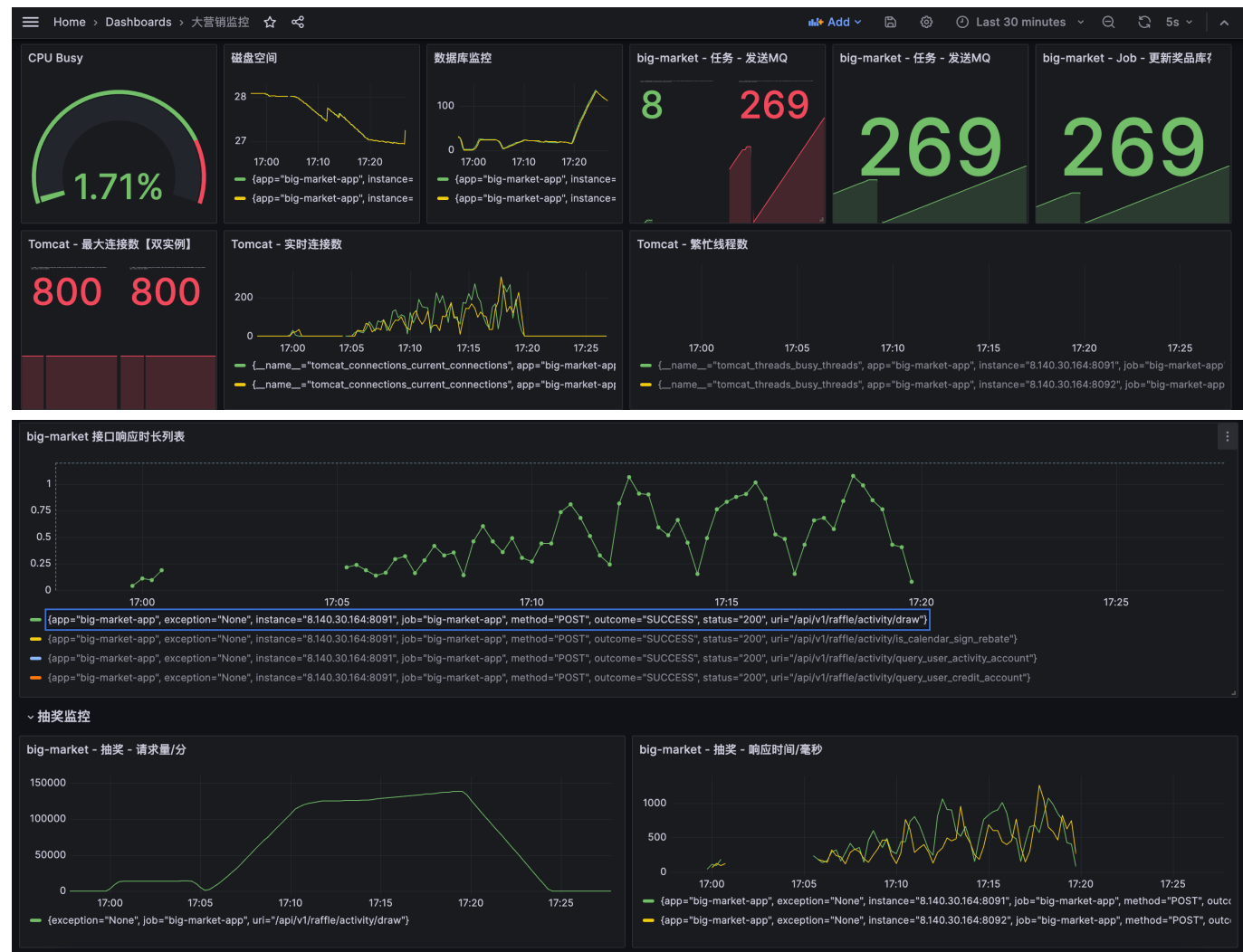
- 时间设置：Ramp-up period: 对应线程数的1/10；
- 测试预估总时长：200ms \* 200 次 \* 8 = 6分（实际用时15分）

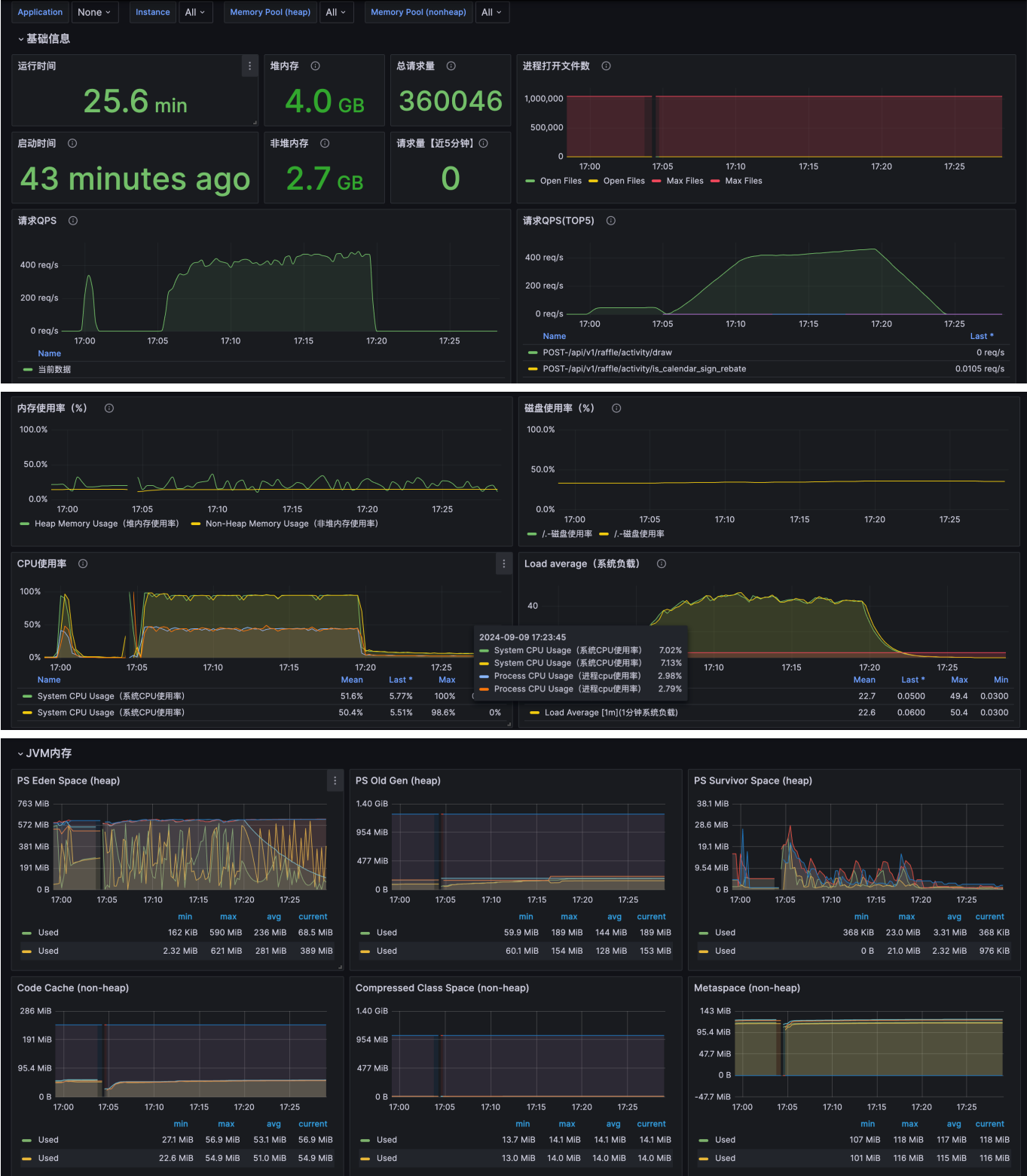
## 05-核心接口的测试结果

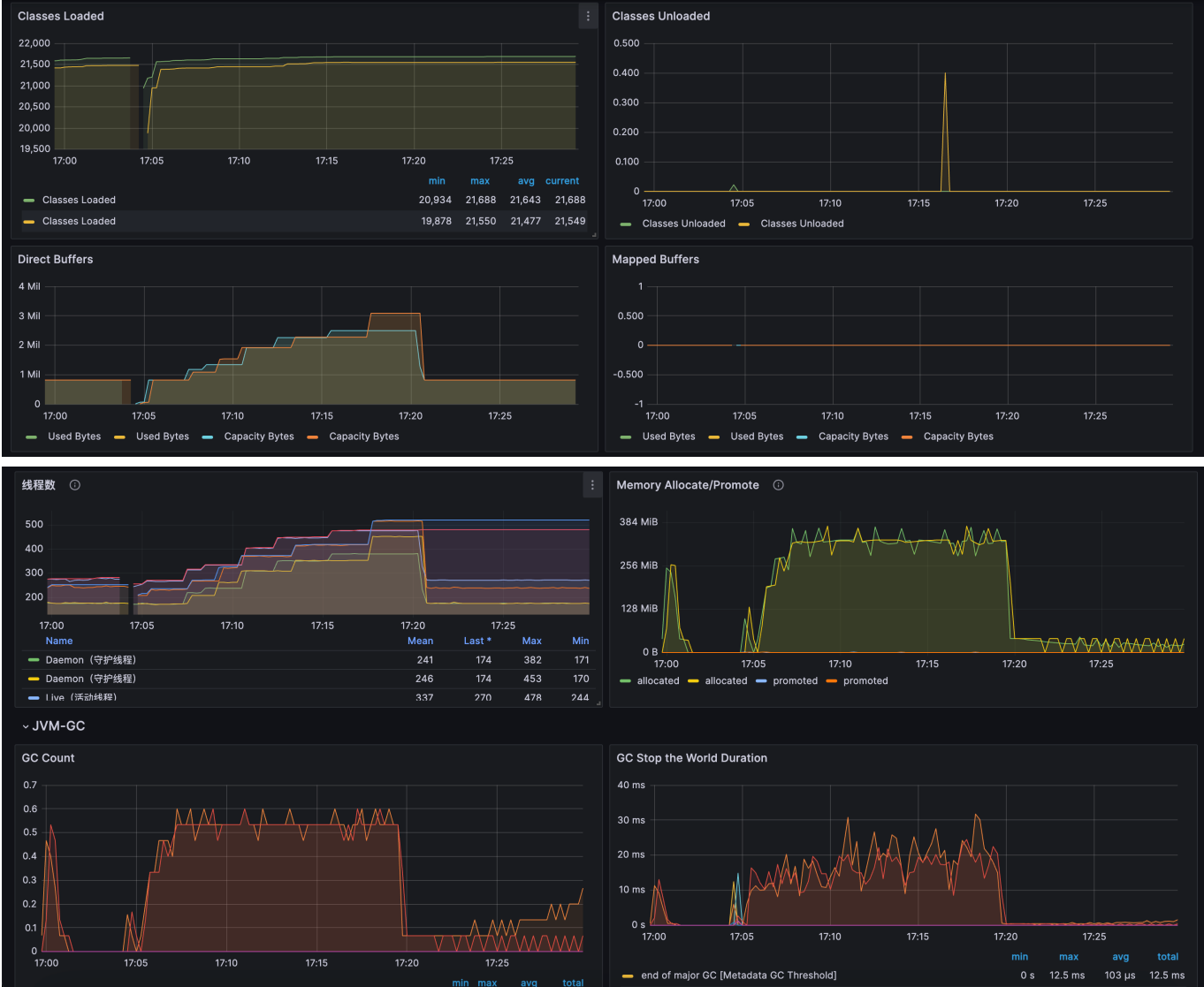
### 5.2 抽奖接口

```
curl --request POST \
--url http://localhost:8091/api/v1/raffle/activity/draw \
--header 'content-type: application/json' \
--data '{
 "userId":"不同用户",
 "activityId": 100301
}'
```

测试TPS：400，RT：300ms左右；











```
[root@iZ2zefck8g31o73sxh0mapZ ~]# jmeter -n -t remote-50-450-200-gt-draw.jmx -l result.jtl
Creating summariser <summary>
Created the tree successfully using remote-50-450-200-gt-draw.jmx
Starting standalone test @ September 9, 2024 5:05:05 PM CST (1725872705915)
Waiting for possible Shutdown/StopTestNow/HeapDump/ThreadDump message on port 4445
summary + 3773 in 00:00:24 = 158.3/s Avg: 248 Min: 63 Max: 1363 Err: 9 (0.24%) Active: 50 Started: 50 Finished: 0
summary + 7856 in 00:00:30 = 261.9/s Avg: 164 Min: 30 Max: 882 Err: 29 (0.37%) Active: 62 Started: 112 Finished: 50
summary = 11629 in 00:00:54 = 216.0/s Avg: 192 Min: 30 Max: 1363 Err: 38 (0.33%)
summary + 10473 in 00:00:30 = 349.1/s Avg: 278 Min: 47 Max: 1051 Err: 54 (0.52%) Active: 100 Started: 150 Finished: 50
summary = 22102 in 00:01:24 = 263.6/s Avg: 233 Min: 30 Max: 1363 Err: 92 (0.42%)
summary + 10481 in 00:00:30 = 349.4/s Avg: 206 Min: 30 Max: 1345 Err: 52 (0.50%) Active: 76 Started: 226 Finished: 150
summary = 32583 in 00:01:54 = 286.2/s Avg: 224 Min: 30 Max: 1363 Err: 144 (0.44%)
summary + 12514 in 00:00:30 = 417.2/s Avg: 336 Min: 49 Max: 1648 Err: 87 (0.70%) Active: 150 Started: 300 Finished: 150
summary = 45097 in 00:02:24 = 313.6/s Avg: 255 Min: 30 Max: 1648 Err: 231 (0.51%)
summary + 12870 in 00:00:30 = 429.0/s Avg: 309 Min: 48 Max: 1035 Err: 71 (0.55%) Active: 88 Started: 300 Finished: 212
summary = 57967 in 00:02:54 = 333.5/s Avg: 267 Min: 30 Max: 1648 Err: 302 (0.52%)
summary + 11931 in 00:00:30 = 397.7/s Avg: 270 Min: 31 Max: 990 Err: 57 (0.48%) Active: 200 Started: 500 Finished: 300
summary = 69898 in 00:03:24 = 342.9/s Avg: 268 Min: 30 Max: 1648 Err: 359 (0.51%)
summary + 12980 in 00:00:30 = 432.6/s Avg: 461 Min: 59 Max: 956 Err: 86 (0.66%) Active: 195 Started: 500 Finished: 305
summary = 82878 in 00:03:54 = 354.4/s Avg: 298 Min: 30 Max: 1648 Err: 445 (0.54%)
summary + 12660 in 00:00:30 = 422.0/s Avg: 418 Min: 31 Max: 2051 Err: 73 (0.58%) Active: 142 Started: 500 Finished: 358
summary = 95538 in 00:04:24 = 362.1/s Avg: 314 Min: 30 Max: 2051 Err: 518 (0.54%)
summary + 11919 in 00:00:30 = 397.3/s Avg: 245 Min: 31 Max: 766 Err: 77 (0.65%) Active: 194 Started: 694 Finished: 500
summary = 107457 in 00:04:54 = 365.7/s Avg: 306 Min: 30 Max: 2051 Err: 595 (0.55%)
summary + 12718 in 00:00:30 = 423.9/s Avg: 575 Min: 46 Max: 2371 Err: 90 (0.71%) Active: 250 Started: 750 Finished: 500
summary = 120175 in 00:05:24 = 371.1/s Avg: 335 Min: 30 Max: 2371 Err: 685 (0.57%)
summary + 13014 in 00:00:30 = 433.8/s Avg: 564 Min: 56 Max: 1649 Err: 99 (0.76%) Active: 235 Started: 750 Finished: 515
summary = 133189 in 00:05:54 = 376.4/s Avg: 357 Min: 30 Max: 2371 Err: 784 (0.59%)
summary + 12753 in 00:00:30 = 425.2/s Avg: 486 Min: 61 Max: 1532 Err: 79 (0.62%) Active: 160 Started: 750 Finished: 590
summary = 145942 in 00:06:24 = 380.2/s Avg: 368 Min: 30 Max: 2371 Err: 863 (0.59%)
summary + 12192 in 00:00:30 = 406.4/s Avg: 253 Min: 29 Max: 710 Err: 59 (0.48%) Active: 204 Started: 954 Finished: 750
summary = 158134 in 00:06:54 = 382.1/s Avg: 359 Min: 29 Max: 2371 Err: 922 (0.58%)
summary + 12228 in 00:00:30 = 407.6/s Avg: 693 Min: 44 Max: 2311 Err: 83 (0.68%) Active: 300 Started: 1050 Finished: 750
summary = 170362 in 00:07:24 = 383.8/s Avg: 383 Min: 29 Max: 2371 Err: 1005 (0.59%)
summary + 12899 in 00:00:30 = 429.9/s Avg: 691 Min: 62 Max: 1259 Err: 100 (0.78%) Active: 290 Started: 1050 Finished: 760
summary = 183261 in 00:07:54 = 386.8/s Avg: 405 Min: 29 Max: 2371 Err: 1105 (0.60%)
summary + 12490 in 00:00:30 = 416.4/s Avg: 664 Min: 49 Max: 3047 Err: 76 (0.61%) Active: 258 Started: 1050 Finished: 792
summary = 145942 in 00:06:24 = 380.2/s Avg: 368 Min: 30 Max: 2371 Err: 863 (0.59%)
summary + 12192 in 00:00:30 = 406.4/s Avg: 253 Min: 29 Max: 710 Err: 59 (0.48%) Active: 204 Started: 954 Finished: 750
summary = 158134 in 00:06:54 = 382.1/s Avg: 359 Min: 29 Max: 2371 Err: 922 (0.58%)
summary + 12228 in 00:00:30 = 407.6/s Avg: 693 Min: 44 Max: 2311 Err: 83 (0.68%) Active: 300 Started: 1050 Finished: 750
summary = 170362 in 00:07:24 = 383.8/s Avg: 383 Min: 29 Max: 2371 Err: 1005 (0.59%)
summary + 12899 in 00:00:30 = 429.9/s Avg: 691 Min: 62 Max: 1259 Err: 100 (0.78%) Active: 290 Started: 1050 Finished: 760
summary = 183261 in 00:07:54 = 386.8/s Avg: 405 Min: 29 Max: 2371 Err: 1105 (0.60%)
summary + 12490 in 00:00:30 = 416.4/s Avg: 664 Min: 49 Max: 3047 Err: 76 (0.61%) Active: 258 Started: 1050 Finished: 792
summary = 195751 in 00:08:24 = 388.5/s Avg: 422 Min: 29 Max: 3047 Err: 1181 (0.60%)
summary + 13742 in 00:00:30 = 458.1/s Avg: 452 Min: 47 Max: 1042 Err: 103 (0.75%) Active: 71 Started: 1050 Finished: 979
summary = 209493 in 00:08:54 = 392.4/s Avg: 424 Min: 29 Max: 3047 Err: 1284 (0.61%)
summary + 12770 in 00:00:30 = 425.7/s Avg: 315 Min: 29 Max: 1151 Err: 86 (0.67%) Active: 286 Started: 1336 Finished: 1050
summary = 222263 in 00:09:24 = 394.2/s Avg: 417 Min: 29 Max: 3047 Err: 1370 (0.62%)
summary + 14019 in 00:00:30 = 467.3/s Avg: 732 Min: 62 Max: 1215 Err: 95 (0.68%) Active: 350 Started: 1400 Finished: 1050
summary = 236282 in 00:09:54 = 397.9/s Avg: 436 Min: 29 Max: 3047 Err: 1465 (0.62%)
summary + 13912 in 00:00:30 = 463.8/s Avg: 738 Min: 71 Max: 1971 Err: 94 (0.68%) Active: 331 Started: 1400 Finished: 1069
summary = 250194 in 00:10:24 = 401.1/s Avg: 453 Min: 29 Max: 3047 Err: 1559 (0.62%)
summary + 14057 in 00:00:30 = 468.6/s Avg: 673 Min: 51 Max: 1381 Err: 113 (0.80%) Active: 291 Started: 1400 Finished: 1109
summary = 264251 in 00:10:54 = 404.2/s Avg: 465 Min: 29 Max: 3047 Err: 1672 (0.63%)
summary + 13697 in 00:00:30 = 456.6/s Avg: 535 Min: 31 Max: 2587 Err: 88 (0.64%) Active: 142 Started: 1400 Finished: 1258
summary = 277948 in 00:11:24 = 406.5/s Avg: 468 Min: 29 Max: 3047 Err: 1760 (0.63%)
summary + 12966 in 00:00:30 = 432.2/s Avg: 279 Min: 29 Max: 976 Err: 71 (0.55%) Active: 255 Started: 1655 Finished: 1400
summary = 290914 in 00:11:54 = 407.5/s Avg: 460 Min: 29 Max: 3047 Err: 1831 (0.63%)
summary + 13888 in 00:00:30 = 462.9/s Avg: 776 Min: 53 Max: 2593 Err: 92 (0.66%) Active: 400 Started: 1800 Finished: 1400
summary = 304802 in 00:12:24 = 409.8/s Avg: 474 Min: 29 Max: 3047 Err: 1923 (0.63%)
summary + 13789 in 00:00:30 = 459.6/s Avg: 866 Min: 90 Max: 2185 Err: 97 (0.70%) Active: 390 Started: 1800 Finished: 1410
summary = 318591 in 00:12:54 = 411.7/s Avg: 491 Min: 29 Max: 3047 Err: 2020 (0.63%)
summary + 14148 in 00:00:30 = 471.6/s Avg: 798 Min: 75 Max: 1414 Err: 100 (0.71%) Active: 366 Started: 1800 Finished: 1434
summary = 332739 in 00:13:24 = 413.9/s Avg: 504 Min: 29 Max: 3047 Err: 2120 (0.64%)
summary + 13979 in 00:00:30 = 466.1/s Avg: 735 Min: 68 Max: 1705 Err: 98 (0.70%) Active: 309 Started: 1800 Finished: 1491
summary = 346718 in 00:13:54 = 415.8/s Avg: 513 Min: 29 Max: 3047 Err: 2218 (0.64%)
summary + 13282 in 00:00:29 = 462.3/s Avg: 494 Min: 29 Max: 1336 Err: 97 (0.73%) Active: 0 Started: 1800 Finished: 1800
summary = 360000 in 00:14:23 = 417.4/s Avg: 513 Min: 29 Max: 3047 Err: 2315 (0.64%)
Tidying up ... @ September 9, 2024 5:19:28 PM CST (1725873568732)
... end of run
[root@iZ2zefck8g31o73sxh0mapZ ~]# docker ps
```

## arthas 耗时排查

1. 启动本地的大营销项目，链接远程数据库；
2. 启动java -jar arthas-boot.jar，输入编号，监控本地项目；格式如下：

```
trace 全限定类名 方法名 --skipJDKMethod false
```



```
benjie@MacBook-Pro ~ % java -jar arthas-boot.jar
[INFO] JAVA_HOME: /Library/Java/JavaVirtualMachines/zulu-8.jdk/Contents/Home/jre
[INFO] arthas-boot version: 3.7.2
[INFO] Found existing java process, please choose one and input the serial number of the process, eg : 1. Then hit ENTER.
* [1]: 15505 /Users/benjie/software/jmeter/bin/ApacheJMeter.jar
 [2]: 15525 org.jetbrains.idea.maven.server.RemoteMavenServer36
 [3]: 14968
 [4]: 26826 org.jetbrains.jps.cmdline.Launcher
 [5]: 26827 com.ben.Application
5
[INFO] arthas home: /Users/benjie/.arthas/lib/3.7.2/arthas
[INFO] Try to attach process 26827
Picked up JAVA_TOOL_OPTIONS:
[INFO] Attach process 26827 success.
[INFO] arthas-client connect 127.0.0.1 3658

ARTHAS

wiki https://arthas.aliyun.com/doc
tutorials https://arthas.aliyun.com/doc/arthas-tutorials.html
version 3.7.2
main_class com.ben.Application
pid 26827
time 2024-09-07 13:20:41

[arthas@26827]$ trace com.ben.trigger.http.RaffleActivityController draw
```

### 1. 发送一次抽奖请求

```
curl --request POST \
--url http://localhost:8091/api/v1/raffle/activity/draw \
--header 'content-type: application/json' \
--data '{
 "userId": "xiaofuge",
 "activityId": 100301
}'
```

### 4. 查看打印log

```
+++[0.00% 0.00471ms] com.ben.trigger.api.dto.ActivityDrawRequestDTO:getId() #177
+++[0.00% 0.005883ms] com.ben.trigger.api.dto.ActivityDrawRequestDTO:getId() #177
+++[19.61% 837.951917ms] com.ben.domain.activity.service.IRaffleActivityPartakeService:createOrder() #177
+++[0.00% 0.037125ms] com.ben.trigger.api.dto.ActivityDrawRequestDTO:getId() #178
+++[0.00% 0.050958ms] com.ben.trigger.api.dto.ActivityDrawRequestDTO:getId() #178
+++[0.00% 0.081625ms] com.ben.domain.activity.model.entity.UserRaffleOrderEntity:getOrderId() #178
+++[0.03% 1.253ms] org.slf4j.Logger:info() #178
+++[0.09% 3.858458ms] com.ben.domain.strategy.model.entity.RaffleFactorEntity:builder() #180
+++[0.00% 0.026375ms] com.ben.domain.activity.model.entity.UserRaffleOrderEntity:getUserId() #181
+++[0.00% 0.028708ms] com.ben.domain.strategy.model.entity.RaffleFactorEntity$RaffleFactorEntityBuilder:userId() #181
+++[0.00% 0.017667ms] com.ben.domain.activity.model.entity.UserRaffleOrderEntity:getStrategyId() #182
+++[0.00% 0.019167ms] com.ben.domain.strategy.model.entity.RaffleFactorEntity$RaffleFactorEntityBuilder:strategyId() #182
+++[0.00% 0.023375ms] com.ben.domain.activity.model.entity.UserRaffleOrderEntity:getEndTime() #183
+++[0.00% 0.046625ms] com.ben.domain.strategy.model.entity.RaffleFactorEntity$RaffleFactorEntityBuilder:build() #184
+++[66.32% 2834.007375ms] com.ben.domain.strategy.service.IRaffleStrategy:performRaffle() #180
+++[0.02% 0.83225ms] com.ben.domain.award.model.entity.UserAwardRecordEntity:builder() #186
+++[0.00% 0.008834ms] com.ben.domain.activity.model.entity.UserRaffleOrderEntity:getUserId() #187
+++[0.00% 0.0145ms] com.ben.domain.award.model.entity.UserAwardRecordEntity$UserAwardRecordEntityBuilder:userId() #187
+++[0.00% 0.010625ms] com.ben.domain.activity.model.entity.UserRaffleOrderEntity:getActivityId() #188
+++[0.00% 0.011542ms] com.ben.domain.award.model.entity.UserAwardRecordEntity$UserAwardRecordEntityBuilder:activityId() #188
+++[0.00% 0.007875ms] com.ben.domain.activity.model.entity.UserRaffleOrderEntity:getStrategyId() #189
+++[0.00% 0.010958ms] com.ben.domain.award.model.entity.UserAwardRecordEntity$UserAwardRecordEntityBuilder:strategyId() #189
+++[0.00% 0.008084ms] com.ben.domain.activity.model.entity.UserRaffleOrderEntity:getOrderId() #190
+++[0.00% 0.011209ms] com.ben.domain.award.model.entity.UserAwardRecordEntity$UserAwardRecordEntityBuilder:orderId() #190
+++[0.00% 0.012084ms] com.ben.domain.strategy.model.entity.RaffleAwardEntity:getAwardId() #191
+++[0.00% 0.01075ms] com.ben.domain.award.model.entity.UserAwardRecordEntity$UserAwardRecordEntityBuilder:awardId() #191
+++[0.00% 0.009875ms] com.ben.domain.strategy.model.entity.RaffleAwardEntity:getAwardTitle() #192
+++[0.00% 0.010875ms] com.ben.domain.award.model.entity.UserAwardRecordEntity$UserAwardRecordEntityBuilder:awardTitle() #192
+++[0.00% 0.0115ms] com.ben.domain.award.model.entity.UserAwardRecordEntity$UserAwardRecordEntityBuilder:awardTime() #193
+++[0.00% 0.012375ms] com.ben.domain.award.model.entity.UserAwardRecordEntity$UserAwardRecordEntityBuilder:awardState() #194
+++[0.00% 0.009541ms] com.ben.domain.strategy.model.entity.RaffleAwardEntity:getAwardConfig() #195
+++[0.00% 0.01125ms] com.ben.domain.award.model.entity.UserAwardRecordEntity$UserAwardRecordEntityBuilder:awardConfig() #195
+++[0.00% 0.02175ms] com.ben.domain.award.model.entity.UserAwardRecordEntity$UserAwardRecordEntityBuilder:build() #196
+++[13.64% 582.988ms] com.ben.domain.award.service.IAwardService:saveUserAwardRecord() #197
```

## trace com.ben.domain.strategy.service.AbstractRaffleStrategy raffleLogicChain

```
[arthas@27252]$ trace com.ben.domain.strategy.service.IRaffleStrategy performRaffle
Press Q or Ctrl+C to abort.
Affect(class count: 3 , method count: 1) cost in 148 ms, listenerId: 2
`--ts=2024-09-07 13:28:06;thread_name=http-nio-8091-exec-2;id=82;is_daemon=true;priority=5;TCCL=org.springframework.boot.web.embedded.tomcat.TomcatEmbeddedWebappClass
Loader@3915e7c3
 |--[366.105375ms] com.ben.domain.strategy.service.AbstractRaffleStrategy:performRaffle()
 |--[0.01% 0.029ms] com.ben.domain.strategy.model.entity.RaffleFactorEntity:getId() #47
 |--[0.00% 0.011083ms] com.ben.domain.strategy.model.entity.RaffleFactorEntity:getStrategyId() #48
 |--[0.01% 0.02075ms] org.apache.commons.lang3.StringUtils:isBlank() #49
 |--[0.04% 0.1445ms] org.slf4j.Logger:info() #52
 |--[86.95% 318.337125ms] com.ben.domain.strategy.service.AbstractRaffleStrategy:raffleLogicChain() #54
 |--[0.02% 0.057292ms] com.ben.domain.strategy.service.rule.chain.factory.DefaultChainFactory$StrategyAwardVO:getAwardId() #55
 |--[0.01% 0.029167ms] com.ben.domain.strategy.service.rule.chain.factory.DefaultChainFactory$StrategyAwardVO:getLogicModel() #55
 |--[0.00% 0.281167ms] org.slf4j.Logger:info() #55
 |--[0.01% 0.032709ms] com.ben.domain.strategy.service.rule.chain.factory.DefaultChainFactory$LogicModel:getCode() #56
 |--[0.02% 0.0605ms] com.ben.domain.strategy.service.rule.chain.factory.DefaultChainFactory$StrategyAwardVO:getLogicModel() #56
 |--[0.01% 0.025542ms] com.ben.domain.strategy.service.rule.chain.factory.DefaultChainFactory$StrategyAwardVO:getAwardId() #57
 |--[0.01% 0.029333ms] com.ben.domain.strategy.service.rule.chain.factory.DefaultChainFactory$StrategyAwardVO:getAwardRuleValue() #57
 |--[12.69% 46.455625ms] com.ben.domain.strategy.service.AbstractRaffleStrategy:buildRaffleAwardEntity() #57
```

## trace com.ben.domain.strategy.service.rule.chain.ILogicChain logic

```
[arthas@27252]$ trace com.ben.domain.strategy.service.AbstractRaffleStrategy raffleLogicChain
Press Q or Ctrl+C to abort.
Affect(class count: 2 , method count: 1) cost in 101 ms, listenerId: 3
`--ts=2024-09-07 13:29:14;thread_name=http-nio-8091-exec-4;id=84;is_daemon=true;priority=5;TCCL=org.springframework.boot.web.embedded.tomcat.TomcatEmbeddedWebappClass
Loader@3915e7c3
 |--[421.763ms] com.ben.domain.strategy.service.raffle.DefaultRaffleStrategy:raffleLogicChain()
 |--[0.03% 0.1135ms] org.slf4j.Logger:info() #42
 |--[0.01% 0.022542ms] com.ben.domain.strategy.service.rule.chain.factory.DefaultChainFactory:openLogicChain() #43
 |--[99.92% 421.428167ms] com.ben.domain.strategy.service.rule.chain.ILogicChain:logic() #44
```

## 继续深入：

```
`--ts=2024-09-07 13:30:38;thread_name=http-nio-8091-exec-8;id=88;is_daemon=true;priority=5;TCCL=org.springframework.boot.web.embedded.tomcat.TomcatEmbeddedWebappClass
Loader@3915e7c3
 |--[295.435584ms] com.ben.domain.strategy.service.rule.chain.impl.BlackListLogicChain:logic()
 |--[0.01% 0.0215ms] com.ben.domain.strategy.service.rule.chain.impl.BlackListLogicChain:ruleModel() #26
 |--[0.48% 1.429125ms] org.slf4j.Logger:info() #26
 |--[0.04% 0.112917ms] com.ben.domain.strategy.service.rule.chain.impl.BlackListLogicChain:ruleModel() #29
 |--[23.44% 69.244625ms] com.ben.domain.strategy.repository.IStrategyRepository:queryStrategyRuleValue() #29
 |--[0.00% 0.012416ms] com.ben.domain.strategy.service.rule.chain.impl.BlackListLogicChain:ruleModel() #49
 |--[0.12% 0.349125ms] org.slf4j.Logger:info() #49
 |--[0.01% 0.021042ms] com.ben.domain.strategy.service.rule.chain.impl.BlackListLogicChain:next() #50
 |--[75.75% 223.782417ms] com.ben.domain.strategy.service.rule.chain.ILogicChain:logic() #50
 |--[99.97% 223.713125ms] com.ben.domain.strategy.service.rule.chain.impl.RuleWeightLogicChain:logic()
 |--[0.01% 0.013708ms] com.ben.domain.strategy.service.rule.chain.impl.RuleWeightLogicChain:ruleModel() #39
 |--[0.03% 0.059708ms] org.slf4j.Logger:info() #39
 |--[0.00% 0.009544ms] com.ben.domain.strategy.service.rule.chain.impl.RuleWeightLogicChain:ruleModel() #42
 |--[14.51% 32.461125ms] com.ben.domain.strategy.repository.IStrategyRepository:queryStrategyRuleValue() #42
 |--[0.03% 0.06475ms] com.ben.domain.strategy.service.rule.chain.impl.RuleWeightLogicChain:getAnalyticalValue() #45
 |--[34.71% 77.654375ms] com.ben.domain.strategy.repository.IStrategyRepository:queryActivityAccountTotalUseCount() #66
 |--[50.39% 112.731542ms] com.ben.domain.strategy.service.armory.IStrategyDispatch:getRandomAwardId() #75
 |--[0.00% 0.009042ms] com.ben.domain.strategy.service.rule.chain.impl.RuleWeightLogicChain:ruleModel() #76
 |--[0.14% 0.313ms] org.slf4j.Logger:info() #76
 |--[0.00% 0.01025ms] com.ben.domain.strategy.service.rule.chain.factory.DefaultChainFactory$StrategyAwardVO:builder() #77
 |--[0.00% 0.008625ms] com.ben.domain.strategy.service.rule.chain.factory.DefaultChainFactory$StrategyAwardVO$StrategyAwardVOBuilder:awardId() #78
 |--[0.00% 0.006125ms] com.ben.domain.strategy.service.rule.chain.impl.RuleWeightLogicChain:ruleModel() #79
 |--[0.00% 0.0085ms] com.ben.domain.strategy.service.rule.chain.factory.DefaultChainFactory$StrategyAwardVO$StrategyAwardVOBuilder:logicModel() #79
 |--[0.00% 0.010584ms] com.ben.domain.strategy.service.rule.chain.factory.DefaultChainFactory$StrategyAwardVO$StrategyAwardVOBuilder:build() #80
```

三个耗时接口： com.ben.domain.strategy.repository.IStrategyRepository:queryStrategyRuleValue()

修改：加缓存

## trace com.ben.domain.strategy.repository.IStrategyRepository queryActivityAccountTotalUseCount

```
Press Q or Ctrl+C to abort.
Affect(class count: 3 , method count: 2) cost in 139 ms, listenerId: 2
`--ts=2024-09-07 13:56:34;thread_name=http-nio-8091-exec-5;id=80;is_daemon=true;priority=5;TCCL=org.springframework.boot.web.embedded.tomcat.TomcatEmbeddedWebappClass
Loader@2e362407
 |--[71.800209ms] com.ben.infrastructure.adapter.repository.StrategyRepository$EnhancerBySpringCGIIB$5748006a:queryActivityAccountTotalUseCount()
 |--[99.87% 71.708958ms] org.springframework.cglib.proxy.MethodInterceptor:intercept()
 |--[99.84% 71.597375ms] com.ben.infrastructure.adapter.repository.StrategyRepository:queryActivityAccountTotalUseCount()
 |--[39.34% 28.166125ms] com.ben.infrastructure.dao.IRaffleActivityDao:queryActivityIdByStrategyId() #368
 |--[0.05% 0.038667ms] com.ben.infrastructure.dao.po.RaffleActivityAccount:builder() #369
 |--[0.04% 0.027625ms] com.ben.infrastructure.dao.po.RaffleActivityAccount$RaffleActivityAccountBuilder:userId() #370
 |--[0.03% 0.0185ms] com.ben.infrastructure.dao.po.RaffleActivityAccount$RaffleActivityAccountBuilder:activityId() #371
 |--[0.03% 0.024292ms] com.ben.infrastructure.dao.po.RaffleActivityAccount$RaffleActivityAccountBuilder:build() #372
 |--[59.82% 42.830417ms] com.ben.infrastructure.dao.IRaffleActivityAccountDao:queryActivityAccountByUserId() #369
 |--[0.05% 0.035208ms] com.ben.infrastructure.dao.po.RaffleActivityAccount:getTotalCount() #376
 |--[0.04% 0.031334ms] com.ben.infrastructure.dao.po.RaffleActivityAccount:getTotalCountSurplus() #376
```

修改：预热的时候加缓存

trace com.ben.domain.strategy.service.armory.IStrategyDispatch getRandomAwardId

```
[arthas@272521]$ trace com.ben.domain.strategy.service.armory.IStrategyDispatch getRandomAwardId
Press Q or Ctrl+C to abort.
Affect(class count: 2 , method count: 3) cost in 108 ms, listenerId: 5
`---ts=2024-09-07 13:33:35;thread_name=http-nio-8091-exec-12;id=92;is_daemon=true;priority=5;TCCL=org.springframework.boot
ssLoader@3915e7c3
---[121.098417ms] com.ben.domain.strategy.service.armory.StrategyArmoryDispatch:getRandomAwardId()
+---[56.48% 68.397625ms] com.ben.domain.strategy.repository.IStrategyRepository:getRateRange() #152
`---[42.49% 51.456ms] com.ben.domain.strategy.repository.IStrategyRepository:getStrategyAwardAssemble() #154
```

修改：既然 Redis 中 getValue() 操作已经可以判断值是否存在，可以跳过 isExists() 的调用。Redis 自带的 GET 操作就可以判断是否存在，如果不存在，返回 null 即可。这样可以减少一次 Redis 请求。

```
@@ -103,10 +103,11 @@ public class StrategyRepository implements IStrategyRepository {
 @Override
 public int getRateRange(String key) {
 String cacheKey = Constants.RedisKey.STRATEGY_RATE_RANGE_KEY + key;
 - if (!redisService.isExists(cacheKey)) {
 + Integer rateRange = redisService.getValue(cacheKey);
 + if (null == rateRange) {
 throw new AppException(UN_ASSEMBLED_STRATEGY_ARMORY.getCode(), cacheKey + Constants.COLON + UN_ASSEM
 }
 - return redisService.getValue(cacheKey);
 + return rateRange;
 }

 @Override
```

## 06-测试结论

问题1: xxl-job-admin本来在中间件那台服务器，发现没法正常连接应用服务；

解决方法：迁移到应用服务器，走docker中的桥接网络； 8.140.30.164是应用服务器ip地址；

```
xxl:
 job:
 accessToken: default_token
 # 注册地址
 admin:
 addresses: http://8.140.30.164:9091/xxl-job-admin
 # 注册执行器
 executor:
 # 执行器注册 [选填]：优先使用该配置作为注册地址，为空时使用内嵌服务 "IP:PORT"
 作为注册地址。从而更灵活的支持容器类型执行器动态IP和动态映射端口问题。
 address:
 appname: big-market-job
 # 执行器IP 配置为本机IP [选填]：默认为空表示自动获取IP，多网卡时可手动设置指定
 IP，该IP不会绑定Host仅作为通讯实用；地址信息用于 "执行器注册" 和 "调度中心请求并触发任
 务";
 ip:
 # 执行器端口号 [选填]：小于等于0则自动获取；默认端口为9999，单机部署多个执行器
 时，注意要配置不同执行器端口；
 port: 9999
 # 执行器运行日志文件存储磁盘路径 [选填]：需要对该路径拥有读写权限；为空则使用默
 认路径；
 logpath: ./data/applogs/xxl-job/jobhandler
 # 执行器日志文件保存天数 [选填]：过期日志自动清理，限制值大于等于3时生效；否
 则，如-1，关闭自动清理功能；
 logretentiondays: 30
```

---

应该有其他方法：如何解决？一台服务器放置两个应用docker，另一台服务器放置xxl-job-admin，如何配置才能正常监听到两个应用的地址。